ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

S.V.ISWARYA

192513032

C02-4

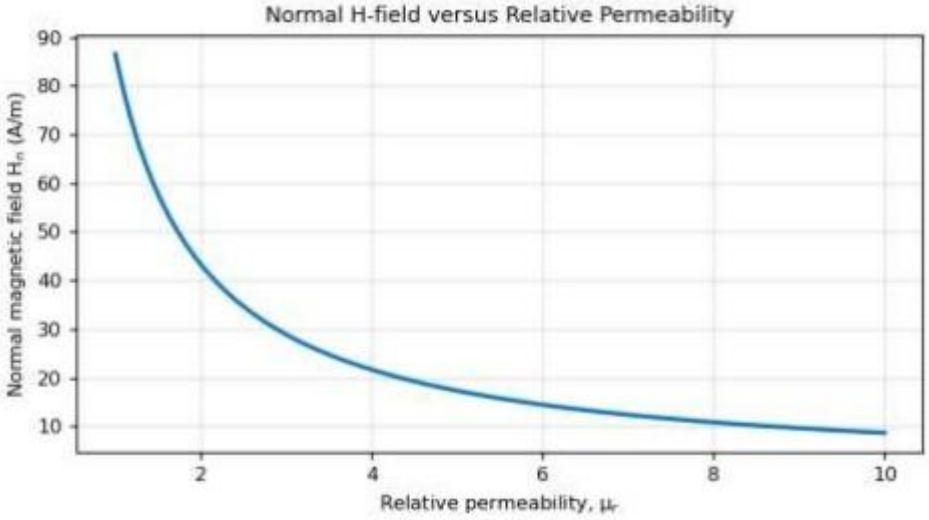
EXPERIMENT 9

AIM:
To simulate a co-axial cable and determine the electric field distribution and capacitance using CST Studio Suite.

Test case (i) Simulation of coaxial cable with given dimensions and dielectric constant
Parameters:
• Inner radius (a) = 1 mm
• Outer radius (b) = 5 mm
• Dielectric constant (εr) = 2

Output:-

• Electric field distribution observed radially
• Maximum field near inner conductor
• Field decreases towards outer conductor



Test case (ii) Effect of dielectric constant on capacitance

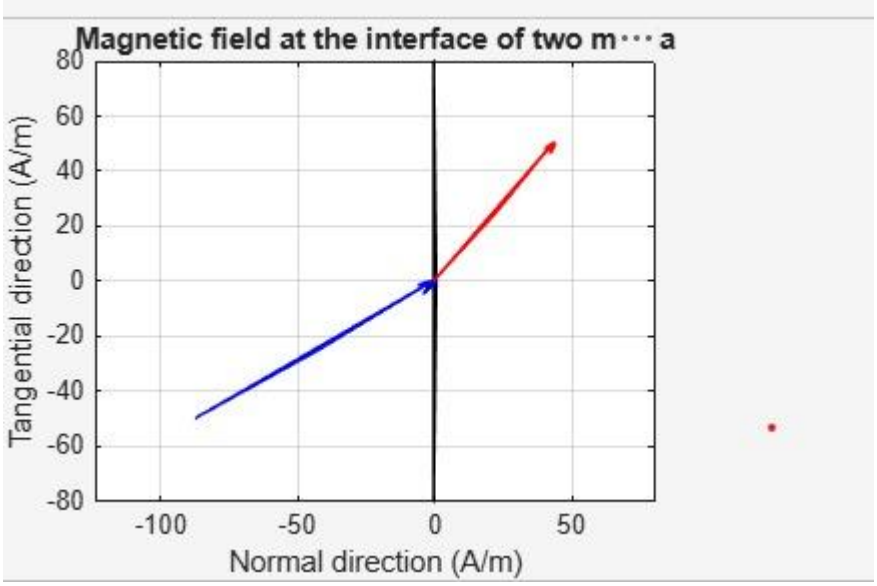
Parameters:

• εr = 1, 2, 4

Output:-

• Capacitance increases with dielectric constant

• Electric field magnitude decreases with higher εr



Test case (iii) Effect of radius variation on capacitance

Parameters:

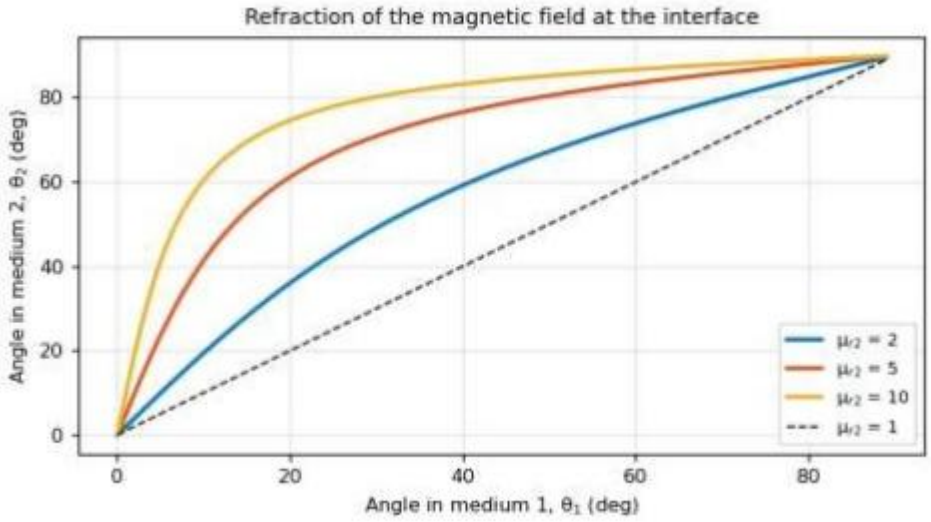
• a = 1 mm (constant)

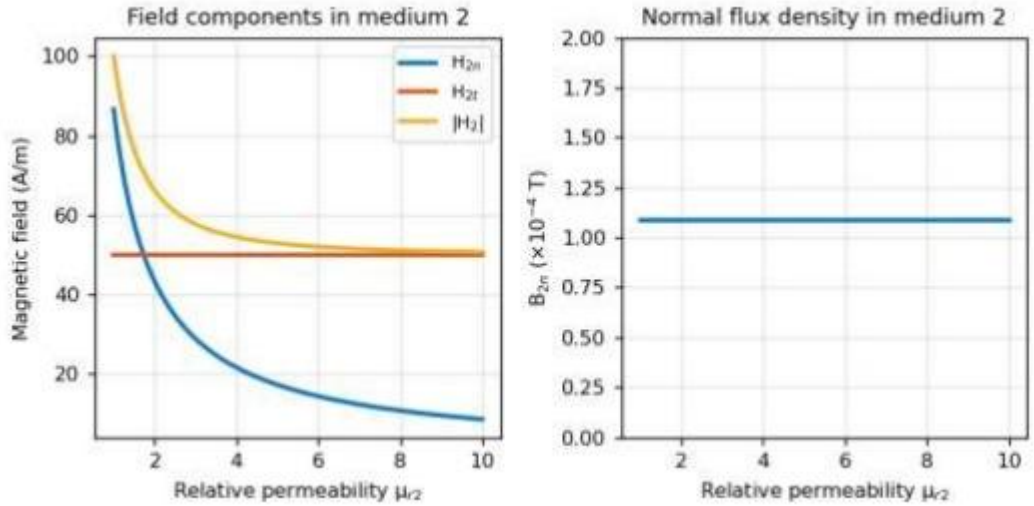
• b = 3 mm, 5 mm, 7 mm

Output:-

• Capacitance decreases as outer radius increases

• Electric field spreads over larger region





Result:

Thus, the simulation of coaxial cable was successfully performed, and electric field distribution and capacitance were verified using CST Studio Suite

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting	Marks given	Marks obtained
Define spherical geometry and dielectric	Define spherical geometry and dielectric	Correctly identifies isolatedsphere radius or inner/outer radii and medium permittivity	Geometry is correct with minor omission.	Some dimensions or material properties are unclear	Uses invalid geometry or material inputs.	25	2.5	
Implement spherical capacitance equations	Implement spherical capacitance equations	Applies the appropriate isolated/concentric sphere formulas correctly.	Formulas are mostly correct with minor errors.	Partial calculations require guidance.	Uses incorrect equations.	25	2.5	
Analyze radius and permittivity effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omissions.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30	3	

Present results and conclusions	Present results and conclusions	Reports geometry, units, calculations and conclusions clearly.	Report is clear with minor gaps	Report lacks some key information.	Results are unclear or missing.	20	2	
TOTAL						100	10	